

In MARS: the date and time of the data is specified with three MARS keywords: date, time and (forecast) step. For analyses, step=0 hours so that date and time specify the analysis date/time. For forecasts, date and time specify the forecast start time and step specifies the number of hours since that start time. The combination of date, time and step defines the validity date/time. For analyses, the validity date/time is equal to the analysis date/time.

In the CDS: analyses are provided rather than forecasts, unless the parameter is only available from the forecasts. The date and time of the data is specified using the validity date/time, so step does not need to be specified. For forecasts, steps between 1 and 12 hours have been used to provide data for all the validity times in each 24 hours, see Table 0 below.

➤ [Table 0: the mapping, for forecasts, between MARS date, time and step and the CDS date and time](#)

Temporal frequency

For sub-daily data for the HRES (stream=oper/wave) the analyses (type=an) are available hourly. The short forecasts, run twice daily from 06 and 18 UTC, provide hourly output forecast steps from 0 to 18 hours (only steps 1 to 12 hours are available on the CDS disks). For the EDA, the sub-daily non-wave data (stream=enda) are available every 3 hours but the sub-daily wave data (stream=ewda) are available hourly in MARS and 3 hourly on the CDS disks.

Spatial grid

The ERA5 HRES atmospheric data has a resolution of 31km, 0.28125 degrees, and the EDA has a resolution of 63km, 0.5625 degrees. (Depending on the parameter, the data are archived either as spectral coefficients with a triangular truncation of T639 (HRES) and T319 (EDA) or on a reduced Gaussian grid with a resolution of N320 (HRES) and N160 (EDA). These grids are so called "linear grids", sometimes referred to as TL639 (HRES) and TL319 (EDA).)

The wave data are produced and archived on a different grid to that of the atmospheric model, namely a reduced latitude/longitude grid with a resolution of 0.36 degrees (HRES) and 1.0 degree (EDA).

ERA5 data available from the CDS disks has been pre-interpolated to a regular latitude/longitude grid appropriate for that data.

The interpolation method is based on the [MIR](#) software. For the production on the Cray HPC (1 January 1940 to 24 October 2022 inclusive) this was an early version of MIR, while for the production on ATOS (25 October 2022 onwards) this is based on the MIR version of the ECMWF MARS client. Differences between both versions are in general small, very localized and not meteorologically significant. For data on pressure levels, differences are mainly limited to the exact north and south pole (90N and 90S). For single-level data, for some fields there are differences at the poles as well, while for some other fields, there are additional sets of isolated points with differences. In both cases this represents an improvement of the interpolation software.

The article [Model grid box and time step](#) might be useful.

Surface elevation datasets used by ERA5

In order to define the surface geopotential in ERA5, the IFS uses surface elevation data interpolated from a combination of SRTM30 and other surface elevation datasets. For more details please see the IFS documentation, Cycle 41r2, [Part IV. Physical processes](#), section 11.2.2 Surface elevation data at 30 arc seconds.

Spatial reference systems and Earth model

The IFS assumes that the underlying shape of the Earth is a perfect sphere, of radius 6371.229 km, with the surface elevation specified relative to that sphere. The geodetic latitude/longitude of the surface elevation datasets are used as if they were the spherical latitude/longitude of the IFS.

ERA5 data is referenced in the horizontal with respect to the WGS84 ellipse (which defines the major/minor axes) and in the vertical it is referenced to the EGM96 geoid over land but over ocean it is referenced to mean sea level, with the approximation that this is assumed to be coincident with the geoid. For more information on the relationship between mean sea level and the geoid, see for example [Gregory et al. \(2019\)](#).

For data in GRIB1 format the earth model is a sphere with radius = 6367.47 km (note, this is inconsistent with what is actually used in the IFS), as defined in the [WMO GRIB Edition 1 specifications](#), Table 7, GDS Octet 17.

For data in GRIB2 format the earth model is a sphere with radius = 6371.2229 km (note, this is consistent with what is actually used in the IFS), as defined in the [WMO GRIB2 specifications](#), section 2.2.1, Code Table 3.2, Code figure 6.

For data in NetCDF format (i.e. converted from the native GRIB format to NetCDF), the earth model is inherited from the GRIB data.

Production experiments

In order to speed up production, the historic ERA5 data was produced by running several parallel experiments which were then spliced together to form the final product.

A discontinuity can occur at the transition between the different experiments. Please see the [Known issues](#) for an example. The degree of discontinuity depends on how well the experiments were "spun-up". How well "spun-up" an experiment is, depends on the initial, chosen, state of the atmosphere and land surface at the beginning of the experiment, how long the experiment is run for, before being used for production, and the parameter(s) of interest - some parameters, such as those for the deeper soil and for the higher atmospheric levels, take longer to spin-up than others.

The information below gives the date ranges for the various production experiments (and hence the transition points) for the final version of ERA5 and also indicates when the computing system changed from the Cray to the ATOS.

➤ [Analysis date ranges for the HRES production experiments](#)

➤ [Analysis date ranges for the EDA production experiments](#)

Note, that forecasts start from the relevant analysis at the forecast start date/time, so the provenance of the whole of each forecast is the same as that of the analysis at the forecast start date/time.

Accuracy and uncertainty

ERA5 is produced using 4D-Var data assimilation and model forecasts in CY41R2 of the IFS. The 4D-Var in ERA5 utilises 12 hour assimilation windows from 9-21 UTC and 21-9 UTC, where the background forecast and all the observations falling within a time window are used to specify all the analyses during that window. However, the accuracy of the analyses is not uniform throughout each window. If the model and observations are unbiased and their errors follow Gaussian distributions and if the observations are homogeneous in space and time, then the analysis error will be smallest in the middle of the assimilation window. However, because none of these assumptions are actually true in the IFS, the particular parameter and location of interest are important, too. Knowing that, a careful study should show at which points during the assimilation windows the analysis is most accurate.

The 10 member ensemble is required for the data assimilation procedure. However, as a useful by-product, this ensemble also provides an estimate of the **relative, random uncertainty**. The "spread" of the 10 member ensemble, encapsulated by the standard deviation, provides a measure of this uncertainty and is larger for time periods and spatial locations where the uncertainty is relatively large and is smaller when and where there is more certainty in the analysed/forecast values. The spread is a measure of the relative uncertainty, so the numbers do not provide the absolute uncertainty. On the whole, the uncertainty becomes larger as you go back in time, when the observing system was not as good as in the present day, and in data sparse locations such as the pre-satellite era, southern hemisphere. In general, apart from that for the sea surface temperature, the spread does not represent systematic uncertainty, only random, or "synoptic", uncertainty. For more information, see [ERA5: uncertainty estimation](#).

Instantaneous parameters

All the analysed parameters and many of the forecast parameters are described as "instantaneous". For more information on what instantaneous means, see [Parameters valid at the specified time](#). Such instantaneous parameters may, or may not, have been averaged in time, to produce monthly means.

Mean rates/fluxes and accumulations

Such parameters, which are only available from forecasts, have undergone particular types of statistical processing (temporal mean or accumulation, respectively) over a period of time called the processing period. In addition, these parameters may, or may not, have been averaged in time, to produce monthly means.

The accumulations (over the accumulation/processing period) in the short forecasts (from 06 and 18 UTC) of ERA5 are treated **differently** compared with those in ERA-Interim and operational data (where the accumulations are from the beginning of the forecast to the validity date/time). In the short forecasts of ERA5, the accumulations are since the previous post processing (archiving), so for:

- reanalysis: accumulations are over the hour (the accumulation/processing period) ending at the validity date/time
- ensemble: accumulations are over the 3 hours (the accumulation/processing period) ending at the validity date/time
- Monthly means (of daily means, stream=moda/edmo): accumulations have been scaled to have an "effective" processing period of one day, see section [Monthly means](#)

Mean rate/flux parameters in ERA5 (e.g. [Table 4](#) for surface and single levels) provide similar information to accumulations (e.g. [Table 3](#) for surface and single levels), except they are expressed as temporal means, over the same processing periods, and so have units of "per second".

- Mean rate/flux parameters are easier to deal with than accumulations because the units do not vary with the processing period.**
- The mean rate hydrological parameters (e.g. the "Mean total precipitation rate") have units of "kg m⁻² s⁻¹", which are equivalent to "mm s⁻¹". They can be multiplied by 86400 seconds (24 hours) to convert to kg m⁻² day⁻¹ or mm day⁻¹.

Note that:

- For the CDS time, or validity time, of 00 UTC, the mean rates/fluxes and accumulations are over the hour (3 hours for the EDA) ending at 00 UTC i.e. the mean or accumulation is during part of the previous day.
- Mean rates/fluxes and accumulations are not available from the analyses.
- Mean rates/fluxes and accumulations at step=0 have values of zero because the length of the processing period is zero.

Minimum/maximum since the previous post processing

The short forecasts of ERA5 contain some surface and single level parameters that are the minimum or maximum value since the previous post processing (archiving), see Table 5 below. So, for:

- reanalysis: the minimum or maximum values are in the hour (the processing period) ending at the validity date/time
- ensemble: the minimum or maximum values are in the 3 hours (the processing period) ending at the validity date/time

Wave spectra

The ocean wave model used in ERA5 (WAM, which is included in the IFS) provides wave spectra with 24 directions and 30 frequencies (see "2D wave spectra (single)", Table 7).

➤ [Decoding 2D wave spectra](#)

Monthly means

In addition to the sub-daily data, most analysed and forecast parameters are also available as monthly means. For the surface and single level parameters, there are some exceptions which are listed in Table 8.

Monthly means are available in two forms:

- Synoptic monthly means, for each particular time and forecast step (stream=mnth/wamo/edmn/ewvmm) - in the CDS, referred to as "monthly averaged by hour of day".
- Monthly means (of daily means, stream=moda/wamd/edmo/ewmo) for the month as a whole - in the CDS, referred to as "monthly averaged". These monthly means are created from all the hourly (3 hourly for the ensemble) data in the month.

Monthly means for:

- forecast parameters are created using the first 12 hours of the twice daily short forecasts (beginning at 06 and 18 UTC).
- analysis and instantaneous forecast parameters** are created from data with a validity time in the month, between 00 and 23 UTC, which excludes the time 00 UTC on the first day of the following month.
- accumulation and mean rate/flux forecast parameters are created from data with processing periods that fall within the month.
- monthly means of daily means, for accumulations and mean rates/fluxes are created from contiguous data with processing periods spanning from 00 UTC on the first day of the month to 00 UTC on the first day of the following month i.e. they are accumulations or mean rates/fluxes for the complete, whole month.

The accumulations in monthly means (of daily means, stream=moda/edmo) have been scaled to have an "effective" processing period of one day, so for accumulations in these streams:

- The hydrological parameters have effective units of "m of water per day" and so they should be multiplied by 1000 to convert to kgm⁻²day⁻¹ or mmday⁻¹.

- The energy (turbulent and radiative) and momentum fluxes should be divided by 86400 seconds (24 hours) to convert to the commonly used units of Wm⁻² and Nm⁻², respectively.

The monthly data in the CDS under 'ERA5 monthly averaged data' has been created by first creating monthly data on the native grid, then regridding this to the lat-lon grid used in the CDS. The hourly data in the CDS under 'ERA5 hourly data' has been created by regridding from the native to the lat-lon grid. Any calculation of monthly means using this hourly data takes place on the already regridded dataset.

In general, the monthly means calculated from the hourly data, or provided in the CDS should be identical, as regridding and averaging are both linear operations. However, when calculating wind speed, there is a nonlinear transformation sqrt(u²+v²) in between and then the order does matter. Therefore, small differences can be seen where the wind fields themselves vary quickly with location, like going from sea to the high volcanos over Hawaii.

Ensemble means and standard deviations

For the EDA sub-daily data (stream=enda/ewda), compared with HRES sub-daily data (stream=oper/wave), ensemble means and standard deviations (type=em/es) are also available. Both these quantities are calculated from all the 10-members (i.e., including the control).

Ensemble standard deviation is often referred to as ensemble spread and is calculated with respect to the ensemble mean. The ensemble standard deviation is not the sample stdv, so we divide by 10 rather than 9 (N-1).

Ensemble means and standard deviations contain analysed parameters when step=0, otherwise they contain forecast parameters. However, only surface and pressure level data (levtype=sfc/pl) contain forecast steps beyond 3 hours. There are no monthly means for ensemble means and standard deviations.

Level listings

Pressure levels (hPa): 1000/975/950/925/900/875/850/825/800/775/750/700/650/600/550/500/450/400/350/300/250/225/200/175/150/125/100/70/50/30/20/10/7/5/3/2/1

Potential temperature levels (K): 265/275/285/300/315/320/330/350/370/395/430/475/530/600/700/850

Potential vorticity level (10⁻⁹ K m² kg⁻¹ s⁻¹ or 10⁻³ PVU): 2000 (which is representative of the dynamical tropopause)

Model levels: 1/10/137, which are described at [L137 model level definitions](#) and [ERA5: compute pressure and geopotential on model levels, geopotential height and geometric height](#). The model levels are hybrid pressure/sigma. For more information, see the documentation of the underlying model, ECMWF's IFS, [CY41R2, Part III. Dynamics and numerical procedures](#), Chapter 2 Basic equations and discretisation.

Parameter listings

Tables 1-6 below describe the surface and single level parameters (levtype=sfc), Table 7 describes wave parameters, Table 8 describes the monthly mean exceptions for surface and single level and wave parameters and Tables 9-13 describe upper air parameters on various levtypes.

Information on all ECMWF parameters (e.g. columns shortName and paramId) is available from the [ECMWF parameter database](#)

⚠ Please note that with the release of the latest version of ecCodes, some shortNames names have changed. As a result, some of the shortNames in the tables below will need to be updated. Until this is completed, please use the relevant entry in the parameter database as official reference for the shortNames and names.

- Table 1: surface and single level parameters: invariants (in time)
- Table 2: surface and single level parameters: instantaneous
- Table 3: surface and single level parameters: accumulations
- Table 4: surface and single level parameters: mean rates/fluxes
- Table 5: surface and single level parameters: minimum/maximum
- Table 6: surface and single level parameters: vertical integrals and total column: instantaneous
- Table 7: wave parameters: instantaneous
- Table 8: monthly mean surface and single level and wave parameters: exceptions from Tables 1-7
- Table 9: pressure level parameters: instantaneous
- Table 10: potential temperature level parameters: instantaneous
- Table 11: potential vorticity level parameters: instantaneous
- Table 12: model level parameters: instantaneous
- Table 13: model level parameters: mean rates/fluxes

Table 1: surface and single level parameters: invariants (in time)

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=sfc)
(The native grid is the reduced Gaussian grid N320 (N160 for the EDA))

カウン ト	名前	ユニット	CDS内の変数名	ショートネーム	パラメータID	一つの	fc
1	Lake cover	(0 - 1)	lake_cover	cl	26	x	x
2	Lake depth	m	lake_depth	dl	228007	x	x
3	Low vegetation cover	(0 - 1)	low_vegetation_cover	cvl	27	x	
4	High vegetation cover	(0 - 1)	high_vegetation_cover	cvh	28	x	
5	Type of low vegetation	~	type_of_low_vegetation	tlv	29	x	
6	Type of high vegetation	~	type_of_high_vegetation	tvh	30	x	
7	Soil type ¹	~	soil_type	slt	43	x	
8	Standard deviation of filtered subgrid orography	m	standard_deviation_of_filtered_subgrid_orography	sdfor	74	x	
9	Geopotential	m**2 s**-2	geopotential	z	129	x	x
10	Standard deviation of sub-gridscale orography	~	standard_deviation_of_orography	sdor	160	x	
11	Anisotropy of sub-gridscale orography	~	anisotropy_of_sub_gridscale_orography	isor	161	x	
12	Angle of sub-gridscale orography	radians	angle_of_sub_gridscale_orography	anor	162	x	
13	Slope of sub-gridscale orography	~	slope_of_sub_gridscale_orography	slor	163	x	
14	Land-sea mask	(0 - 1)	land_sea_mask	lsm	172	x	x

¹Soil type (texture) determines the saturation, field capacity and permanent wilting point at all the soil levels, see Table 8.9 in Chapter 8 Surface parametrization, Part IV Physical Processes of the [IFS documentation](#) (CY41R2 for ERA5).

Table 2: surface and single level parameters: instantaneous

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=sfc)
(The native grid is the reduced Gaussian grid N320 (N160 for the EDA))

カウン ト	名前	ユニット	CDS内の変数名	ショートネーム	パラメータID	一つの	fc
1	Convective inhibition	J kg**-1	convective_inhibition	cin	228001		x
2	Friction velocity	m s**-1	friction_velocity	zust	228003		x
3	Lake mix-layer temperature	K	lake_mix_layer_temperature	lmit	228008	x	x
4	Lake mix-layer depth	m	lake_mix_layer_depth	lmld	228009	x	x
5	Lake bottom temperature	K	lake_bottom_temperature	lbit	228010	x	x
6	Lake total layer temperature	K	lake_total_layer_temperature	ltlt	228011	x	x
7	Lake shape factor	dimensionless	lake_shape_factor	lshf	228012	x	x
8	Lake ice temperature	K	lake_ice_temperature	licf	228013	x	x
9	Lake ice depth	m	lake_ice_depth	licd	228014	x	x
10	UV visible albedo for direct radiation	(0 - 1)	uv_visible_albedo_for_direct_radiation	aluvp	15	x	x
11	Minimum vertical gradient of refractivity inside trapping layer	m**-1	minimum_vertical_gradient_of_refractivity_inside_trapping_layer	dndzn	228015		x
12	UV visible albedo for diffuse radiation	(0 - 1)	uv_visible_albedo_for_diffuse_radiation	aluvd	16	x	x
13	Mean vertical gradient of refractivity inside trapping layer	m**-1	mean_vertical_gradient_of_refractivity_inside_trapping_layer	dndza	228016		x
14	Near IR albedo for direct radiation	(0 - 1)	near_ir_albedo_for_direct_radiation	alnip	17	x	x
15	Duct base height	m	duct_base_height	dctb	228017		x
16	Near IR albedo for diffuse radiation	(0 - 1)	near_ir_albedo_for_diffuse_radiation	alnld	18	x	x
17	Trapping layer base height	m	trapping_layer_base_height	tplb	228018		x
18	Trapping layer top height	m	trapping_layer_top_height	tplt	228019		x
19	Cloud base height	m	cloud_base_height	cbh	228023		x
20	Zero degree level	m	zero_degree_level	deg0l	228024		x
21	Instantaneous 10 metre wind gust	m s**-1	instantaneous_10m_wind_gust	i10fg	228029		x
22	Sea ice area fraction	(0 - 1)	sea-ice_cover	ci	31	x	x
23	Snow albedo	(0 - 1)	snow_albedo	asn	32	x	x

24	Snow density	kg m ^{**-3}	snow_density	rsn	33	x	x
25	Sea surface temperature	K	sea_surface_temperature	sst	34	x	x
26	Ice temperature layer 1	K	ice_temperature_layer_1	istl1	35	x	x
27	Ice temperature layer 2	K	ice_temperature_layer_2	istl2	36	x	x
28	Ice temperature layer 3	K	ice_temperature_layer_3	istl3	37	x	x
29	Ice temperature layer 4	K	ice_temperature_layer_4	istl4	38	x	x
30	Volumetric soil water layer 1 ¹	m ^{**3} m ^{**-3}	volumetric_soil_water_layer_1	swvl1	39	x	x
31	Volumetric soil water layer 2 ¹	m ^{**3} m ^{**-3}	volumetric_soil_water_layer_2	swvl2	40	x	x
32	Volumetric soil water layer 3 ¹	m ^{**3} m ^{**-3}	volumetric_soil_water_layer_3	swvl3	41	x	x
33	Volumetric soil water layer 4 ¹	m ^{**3} m ^{**-3}	volumetric_soil_water_layer_4	swvl4	42	x	x
34	Convective available potential energy	J kg ^{**-1}	convective_available_potential_energy	cape	59	x	x
35	Leaf area index, low vegetation ³	m ^{**2} m ^{**-2}	leaf_area_index_low_vegetation	lai_lv	66	x	x
36	Leaf area index, high vegetation ³	m ^{**2} m ^{**-2}	leaf_area_index_high_vegetation	lai_hv	67	x	x
37	Neutral wind at 10 m u-component	m s ^{**-1}	10m_u-component_of_neutral_wind	u10n	228131	x	x
38	Neutral wind at 10 m v-component	m s ^{**-1}	10m_v-component_of_neutral_wind	v10n	228132	x	x
39	Surface pressure	Pa	surface_pressure	sp	134	x	x
40	Soil temperature level 1 ¹	K	soil_temperature_level_1	stl1	139	x	x
41	Snow depth	m of water equivalent	snow_depth	sd	141	x	x
42	Charnock	~	charnock	chnk	148	x	x
43	Mean sea level pressure	Pa	mean_sea_level_pressure	msl	151	x	x
44	Boundary layer height	m	boundary_layer_height	blh	159	x	x
45	Total cloud cover	(0 - 1)	total_cloud_cover	tcc	164	x	x
46	10 metre U wind component	m s ^{**-1}	10m_u-component_of_wind	10u	165	x	x
47	10 metre V wind component	m s ^{**-1}	10m_v-component_of_wind	10v	166	x	x
48	2 metre temperature	K	2m_temperature	2t	167	x	x
49	2 metre dewpoint temperature	K	2m_dewpoint_temperature	2d	168	x	x
50	Soil temperature level 2 ¹	K	soil_temperature_level_2	stl2	170	x	x
51	Soil temperature level 3 ¹	K	soil_temperature_level_3	stl3	183	x	x
52	Low cloud cover	(0 - 1)	low_cloud_cover	lcc	186	x	x
53	Medium cloud cover	(0 - 1)	medium_cloud_cover	mcc	187	x	x
54	High cloud cover	(0 - 1)	high_cloud_cover	hcc	188	x	x
55	Skin reservoir content	m of water equivalent	skin_reservoir_content	src	198	x	x
56	Instantaneous large-scale surface precipitation fraction	(0 - 1)	instantaneous_large_scale_surface_precipitation_fraction	lspf	228217		x
57	Convective rain rate	kg m ^{**-2} s ^{**-1}	convective_rain_rate	crr	228218		x
58	Large scale rain rate	kg m ^{**-2} s ^{**-1}	large_scale_rain_rate	lsrr	228219		x
59	Convective snowfall rate water equivalent	kg m ^{**-2} s ^{**-1}	convective_snowfall_rate_water_equivalent	csfr	228220		x
60	Large scale snowfall rate water equivalent	kg m ^{**-2} s ^{**-1}	large_scale_snowfall_rate_water_equivalent	lsfr	228221		x
61	Instantaneous eastward turbulent surface stress	N m ^{**-2}	instantaneous_eastward_turbulent_surface_stress	iews	229	x	x
62	Instantaneous northward turbulent surface stress	N m ^{**-2}	instantaneous_northward_turbulent_surface_stress	inss	230	x	x
63	Instantaneous surface sensible heat flux	W m ^{**-2}	instantaneous_surface_sensible_heat_flux	ishf	231	x	x
64	Instantaneous moisture flux	kg m ^{**-2} s ^{**-1}	instantaneous_moisture_flux	ie	232	x	x
65	Skin temperature	K	skin_temperature	skt	235	x	x
66	Soil temperature level 4 ¹	K	soil_temperature_level_4	stl4	236	x	x
67	Temperature of snow layer	K	temperature_of_snow_layer	tsn	238	x	x
68	Forecast albedo	(0 - 1)	forecast_albedo	fal	243	x	x
69	Forecast surface roughness	m	forecast_surface_roughness	fsr	244	x	x
70	Forecast logarithm of surface roughness for heat	~	forecast_logarithm_of_surface_roughness_for_heat	flsr	245	x	x
71	100 metre U wind component	m s ^{**-1}	100m_u-component_of_wind	100u	228246	x	x
72	100 metre V wind component	m s ^{**-1}	100m_v-component_of_wind	100v	228247	x	x
73	Precipitation type ²	code table (4.201)	precipitation_type	ptype	260015		x
74	K index ²	K	k_index	kx	260121		x
75	Total totals index ²	K	total_totals_index	totalx	260123		x

➤ 1 Soil layers

²GRIB2 format

³**Leaf Area Index (LAI)** parameters are based on a monthly climatology. Users will only see monthly variability, but not inter-annual variability.

Table 3: surface and single level parameters: accumulations

(stream=oper/enda/mnth/moda/edmn/edmo, levtype=sfc)
(The native grid is the reduced Gaussian grid N320 (N160 for the EDA))

カウント	名前	ユニット	CDS内の変数名	ショートネーム	パラメータID	一つの	fc
1	Large-scale precipitation fraction	s	large_scale_precipitation_fraction	lspf	50		x
2	Downward UV radiation at the surface	J m ^{**-2}	downward_uv_radiation_at_the_surface	uvb	57		x
3	Boundary layer dissipation	J m ^{**-2}	boundary_layer_dissipation	bld	145		x
4	Surface sensible heat flux	J m ^{**-2}	surface_sensible_heat_flux	sshf	146		x
5	Surface latent heat flux	J m ^{**-2}	surface_latent_heat_flux	slhf	147		x
6	Surface solar radiation downwards	J m ^{**-2}	surface_solar_radiation_downwards	ssrd	169		x
7	Surface thermal radiation downwards	J m ^{**-2}	surface_thermal_radiation_downwards	strd	175		x
8	Surface net solar radiation	J m ^{**-2}	surface_net_solar_radiation	ssr	176		x
9	Surface net thermal radiation	J m ^{**-2}	surface_net_thermal_radiation	str	177		x
10	Top net solar radiation	J m ^{**-2}	top_net_solar_radiation	tsr	178		x
11	Top net thermal radiation	J m ^{**-2}	top_net_thermal_radiation	ttr	179		x
12	Eastward turbulent surface stress	N m ^{**-2} s	eastward_turbulent_surface_stress	ewss	180		x
13	Northward turbulent surface stress	N m ^{**-2} s	northward_turbulent_surface_stress	nsst	181		x
14	Eastward gravity wave surface stress	N m ^{**-2} s	eastward_gravity_wave_surface_stress	lgws	195		x
15	Northward gravity wave surface stress	N m ^{**-2} s	northward_gravity_wave_surface_stress	mgws	196		x
16	Gravity wave dissipation	J m ^{**-2}	gravity_wave_dissipation	gwd	197		x
17	Top net solar radiation, clear sky	J m ^{**-2}	top_net_solar_radiation_clear_sky	tsrc	208		x
18	Top net thermal radiation, clear sky	J m ^{**-2}	top_net_thermal_radiation_clear_sky	trrc	209		x
19	Surface net solar radiation, clear sky	J m ^{**-2}	surface_net_solar_radiation_clear_sky	ssrc	210		x
20	Surface net thermal radiation, clear sky	J m ^{**-2}	surface_net_thermal_radiation_clear_sky	strc	211		x

21	TOA incident solar radiation	J m ^{**-2}	toa_incident_solar_radiation	tisr	212		x
22	Vertically integrated moisture divergence	kg m ^{**-2}	vertically_integrated_moisture_divergence	vimd	213		x
23	Total sky direct solar radiation at surface	J m ^{**-2}	total_sky_direct_solar_radiation_at_surface	fdir	228021		x
24	Clear-sky direct solar radiation at surface	J m ^{**-2}	clear_sky_direct_solar_radiation_at_surface	cdir	228022		x
25	Surface solar radiation downward clear-sky	J m ^{**-2}	surface_solar_radiation_downward_clear_sky	ssrdc	228129		x
26	Surface thermal radiation downward clear-sky	J m ^{**-2}	surface_thermal_radiation_downward_clear_sky	strdc	228130		x
27	Surface runoff	m	surface_runoff	sro	8		x
28	Sub-surface runoff	m	sub_surface_runoff	ssro	9		x
29	Snow evaporation	m of water equivalent	snow_evaporation	es	44		x
30	Snowmelt	m of water equivalent	snowmelt	smit	45		x
31	Large-scale precipitation	m	large_scale_precipitation	lsp	142		x
32	Convective precipitation	m	convective_precipitation	cp	143		x
33	Snowfall	m of water equivalent	snowfall	sf	144		x
34	Evaporation	m of water equivalent	evaporation	e	182		x
35	Runoff	m	runoff	ro	205		x
36	Total precipitation	m	total_precipitation	tp	228		x
37	Convective snowfall	m of water equivalent	convective_snowfall	csf	239		x
38	Large-scale snowfall	m of water equivalent	large_scale_snowfall	lsf	240		x
39	Potential evaporation	m	potential_evaporation	pev	228251		x

The [accumulations](#) in **monthly means of daily means** ([stream=moda/edmo](#)), see [monthly means](#), have been scaled to have units that include "per day", **so for accumulations in these streams:**

- Most hydrological parameters are in units of "m of water per day", so these should be multiplied by 1000 to convert to kg m⁻² day⁻¹ or mm day⁻¹.
- Energy (turbulent and radiative) and momentum fluxes should be divided by 86400 seconds (24 hours) to convert to the commonly used units of W m⁻² and N m⁻², respectively.

Table 4: surface and single level parameters: mean rates/fluxes

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=sfc)

(The native grid is the reduced Gaussian grid N320 (N160 for the EDA))

カウン ト	名前	ユニ ット	CDS内の変数名	ショ ートネ ーム	パラメー タID	一つ の	fc
1	Mean surface runoff rate	kg m ^{**-2} s ^{**-1}	mean_surface_runoff_rate	msror	235020		x
2	Mean sub-surface runoff rate	kg m ^{**-2} s ^{**-1}	mean_sub_surface_runoff_rate	mssror	235021		x
3	Mean snow evaporation rate	kg m ^{**-2} s ^{**-1}	mean_snow_evaporation_rate	mser	235023		x
4	Mean snowmelt rate	kg m ^{**-2} s ^{**-1}	mean_snowmelt_rate	msmr	235024		x
5	Mean large-scale precipitation fraction	Proportion	mean_large_scale_precipitation_fraction	mlspf	235026		x
6	Mean surface downward UV radiation flux	W m ^{**-2}	mean_surface_downward_uv_radiation_flux	msdwuvrf	235027		x
7	Mean large-scale precipitation rate	kg m ^{**-2} s ^{**-1}	mean_large_scale_precipitation_rate	mlspr	235029		x
8	Mean convective precipitation rate	kg m ^{**-2} s ^{**-1}	mean_convective_precipitation_rate	mcpr	235030		x
9	Mean snowfall rate	kg m ^{**-2} s ^{**-1}	mean_snowfall_rate	msr	235031		x
10	Mean boundary layer dissipation	W m ^{**-2}	mean_boundary_layer_dissipation	mbld	235032		x
11	Mean surface sensible heat flux	W m ^{**-2}	mean_surface_sensible_heat_flux	msshf	235033		x
12	Mean surface latent heat flux	W m ^{**-2}	mean_surface_latent_heat_flux	mslhf	235034		x
13	Mean surface downward short-wave radiation flux	W m ^{**-2}	mean_surface_downward_short_wave_radiation_flux	msdswrf	235035		x
14	Mean surface downward long-wave radiation flux	W m ^{**-2}	mean_surface_downward_long_wave_radiation_flux	msdlwrf	235036		x
15	Mean surface net short-wave radiation flux	W m ^{**-2}	mean_surface_net_short_wave_radiation_flux	msnswrf	235037		x
16	Mean surface net long-wave radiation flux	W m ^{**-2}	mean_surface_net_long_wave_radiation_flux	msnlwrf	235038		x
17	Mean top net short-wave radiation flux	W m ^{**-2}	mean_top_net_short_wave_radiation_flux	mtnswrf	235039		x
18	Mean top net long-wave radiation flux	W m ^{**-2}	mean_top_net_long_wave_radiation_flux	mtnlwrf	235040		x
19	Mean eastward turbulent surface stress	N m ^{**-2}	mean_eastward_turbulent_surface_stress	metss	235041		x
20	Mean northward turbulent surface stress	N m ^{**-2}	mean_northward_turbulent_surface_stress	mntss	235042		x
21	Mean evaporation rate	kg m ^{**-2} s ^{**-1}	mean_evaporation_rate	mer	235043		x
22	Mean eastward gravity wave surface stress	N m ^{**-2}	mean_eastward_gravity_wave_surface_stress	megwss	235045		x
23	Mean northward gravity wave surface stress	N m ^{**-2}	mean_northward_gravity_wave_surface_stress	mngwss	235046		x
24	Mean gravity wave dissipation	W m ^{**-2}	mean_gravity_wave_dissipation	mgwd	235047		x
25	Mean runoff rate	kg m ^{**-2} s ^{**-1}	mean_runoff_rate	mrro	235048		x
26	Mean top net short-wave radiation flux, clear sky	W m ^{**-2}	mean_top_net_short_wave_radiation_flux_clear_sky	mtnswrfcs	235049		x
27	Mean top net long-wave radiation flux, clear sky	W m ^{**-2}	mean_top_net_long_wave_radiation_flux_clear_sky	mtnlwrfcs	235050		x
28	Mean surface net short-wave radiation flux, clear sky	W m ^{**-2}	mean_surface_net_short_wave_radiation_flux_clear_sky	msnswrfcs	235051		x
29	Mean surface net long-wave radiation flux, clear sky	W m ^{**-2}	mean_surface_net_long_wave_radiation_flux_clear_sky	msnlwrfcs	235052		x
30	Mean top downward short-wave radiation flux	W m ^{**-2}	mean_top_downward_short_wave_radiation_flux	mtdswrf	235053		x
31	Mean vertically integrated moisture divergence	kg m ^{**-2} s ^{**-1}	mean_vertically_integrated_moisture_divergence	mvimd	235054		x
32	Mean total precipitation rate	kg m ^{**-2} s ^{**-1}	mean_total_precipitation_rate	mtpr	235055		x
33	Mean convective snowfall rate	kg m ^{**-2} s ^{**-1}	mean_convective_snowfall_rate	mcsr	235056		x
34	Mean large-scale snowfall rate	kg m ^{**-2} s ^{**-1}	mean_large_scale_snowfall_rate	mlssr	235057		x
35	Mean surface direct short-wave radiation flux	W m ^{**-2}	mean_surface_direct_short_wave_radiation_flux	msdrswrf	235058		x
36	Mean surface direct short-wave radiation flux, clear sky	W m ^{**-2}	mean_surface_direct_short_wave_radiation_flux_clear_sky	msdrswrfcs	235059		x
37	Mean surface downward short-wave radiation flux, clear sky	W m ^{**-2}	mean_surface_downward_short_wave_radiation_flux_clear_sky	msdswrfcs	235068		x
38	Mean surface downward long-wave radiation flux, clear sky	W m ^{**-2}	mean_surface_downward_long_wave_radiation_flux_clear_sky	msdlwrfcs	235069		x
39	Mean potential evaporation rate	kg m ^{**-2} s ^{**-1}	mean_potential_evaporation_rate	mper	235070		x

The [mean rates/fluxes](#) in Table 4 provide similar information to the accumulations in Table 3, except they are expressed as temporal averages, and so have units of "per second". The mean rate hydrological parameters have units of "kg m⁻² s^{-1a}" and so they can be multiplied by 86400 seconds (24 hours) to convert to kg m⁻² day⁻¹ or mm day⁻¹.

Table 5: surface and single level parameters: minimum/maximum

(stream=oper/enda, levtype=sfc)

(The native grid is the reduced Gaussian grid N320 (N160 for the EDA))

カウン ト	名前	ユニ ット	CDS内の変数名	ショ ートネ ーム	パラメー タID	一つ の	fc
1	10 metre wind gust since previous post-processing	m s ^{**-1}	10m_wind_gust_since_previous_post_processing	10fg	49		x
2	Maximum temperature at 2 metres since previous post-	K	maximum_2m_temperature_since_previous_post_processing	mx2t	201		x

	processing						
3	Minimum temperature at 2 metres since previous post-processing	K	minimum_2m_temperature_since_previous_post_processing	mn2t	202		x
4	Maximum total precipitation rate since previous post-processing	kg m** ⁻² s** ⁻¹	maximum_total_precipitation_rate_since_previous_post_processing	mxtpr	228226		x
5	Minimum total precipitation rate since previous post-processing	kg m** ⁻² s** ⁻¹	minimum_total_precipitation_rate_since_previous_post_processing	mntpr	228227		x

Table 6: surface and single level parameters: vertical integrals and total column: instantaneous

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=sfc - vertical integrals not available for type=en/es
(The native grid is the reduced Gaussian grid N320 (N160 for the EDA))

count	name	units	Variable name in CDS	shortName	paramid	an	fc
1	Vertical integral of mass of atmosphere	kg m** ⁻²	vertical_integral_of_mass_of_atmosphere	vima	162053	x	x
2	Vertical integral of temperature	K kg m** ⁻²	vertical_integral_of_temperature	vit	162054	x	x
3	Vertical integral of kinetic energy	J m** ⁻²	vertical_integral_of_kinetic_energy	vike	162059	x	x
4	Vertical integral of thermal energy	J m** ⁻²	vertical_integral_of_thermal_energy	vithe	162060	x	x
5	Vertical integral of potential+internal energy	J m** ⁻²	vertical_integral_of_potential_and_internal_energy	vipie	162061	x	x
6	Vertical integral of potential+internal+latent energy	J m** ⁻²	vertical_integral_of_potential_internal_and_latent_energy	vipile	162062	x	x
7	Vertical integral of total energy	J m** ⁻²	vertical_integral_of_total_energy	vitoe	162063	x	x
8	Vertical integral of energy conversion	W m** ⁻²	vertical_integral_of_energy_conversion	vi ec	162064	x	x
9	Vertical integral of eastward mass flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_eastward_mass_flux	vima e	162065	x	x
10	Vertical integral of northward mass flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_northward_mass_flux	vima n	162066	x	x
11	Vertical integral of eastward kinetic energy flux	W m** ⁻¹	vertical_integral_of_eastward_kinetic_energy_flux	vike e	162067	x	x
12	Vertical integral of northward kinetic energy flux	W m** ⁻¹	vertical_integral_of_northward_kinetic_energy_flux	vike n	162068	x	x
13	Vertical integral of eastward heat flux	W m** ⁻¹	vertical_integral_of_eastward_heat_flux	vithe e	162069	x	x
14	Vertical integral of northward heat flux	W m** ⁻¹	vertical_integral_of_northward_heat_flux	vithe n	162070	x	x
15	Vertical integral of eastward water vapour flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_eastward_water_vapour_flux	viwve	162071	x	x
16	Vertical integral of northward water vapour flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_northward_water_vapour_flux	viwvn	162072	x	x
17	Vertical integral of eastward geopotential flux	W m** ⁻¹	vertical_integral_of_eastward_geopotential_flux	vige	162073	x	x
18	Vertical integral of northward geopotential flux	W m** ⁻¹	vertical_integral_of_northward_geopotential_flux	vign	162074	x	x
19	Vertical integral of eastward total energy flux	W m** ⁻¹	vertical_integral_of_eastward_total_energy_flux	vitoe e	162075	x	x
20	Vertical integral of northward total energy flux	W m** ⁻¹	vertical_integral_of_northward_total_energy_flux	vitoe n	162076	x	x
21	Vertical integral of eastward ozone flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_eastward_ozone_flux	vioze	162077	x	x
22	Vertical integral of northward ozone flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_northward_ozone_flux	viozn	162078	x	x
23	Vertical integral of divergence of cloud liquid water flux	kg m** ⁻² s** ⁻¹	vertical_integral_of_divergence_of_cloud_liquid_water_flux	vilwd	162079	x	x
24	Vertical integral of divergence of cloud frozen water flux	kg m** ⁻² s** ⁻¹	vertical_integral_of_divergence_of_cloud_frozen_water_flux	vilwd	162080	x	x
25	Vertical integral of divergence of mass flux	kg m** ⁻² s** ⁻¹	vertical_integral_of_divergence_of_mass_flux	vimad	162081	x	x
26	Vertical integral of divergence of kinetic energy flux	W m** ⁻²	vertical_integral_of_divergence_of_kinetic_energy_flux	viked	162082	x	x
27	Vertical integral of divergence of thermal energy flux	W m** ⁻²	vertical_integral_of_divergence_of_thermal_energy_flux	vithed	162083	x	x
28	Vertical integral of divergence of moisture flux	kg m** ⁻² s** ⁻¹	vertical_integral_of_divergence_of_moisture_flux	viwvd	162084	x	x
29	Vertical integral of divergence of geopotential flux	W m** ⁻²	vertical_integral_of_divergence_of_geopotential_flux	vigd	162085	x	x
30	Vertical integral of divergence of total energy flux	W m** ⁻²	vertical_integral_of_divergence_of_total_energy_flux	vitoe d	162086	x	x
31	Vertical integral of divergence of ozone flux	kg m** ⁻² s** ⁻¹	vertical_integral_of_divergence_of_ozone_flux	viozd	162087	x	x
32	Vertical integral of eastward cloud liquid water flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_eastward_cloud_liquid_water_flux	vilwe	162088	x	x
33	Vertical integral of northward cloud liquid water flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_northward_cloud_liquid_water_flux	vilwn	162089	x	x
34	Vertical integral of eastward cloud frozen water flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_eastward_cloud_frozen_water_flux	vilwe	162090	x	x
35	Vertical integral of northward cloud frozen water flux	kg m** ⁻¹ s** ⁻¹	vertical_integral_of_northward_cloud_frozen_water_flux	vilwn	162091	x	x
36	Vertical integral of mass tendency	kg m** ⁻² s** ⁻¹	vertical_integral_of_mass_tendency	vimat	162092	x	
37	Total column cloud liquid water	kg m** ⁻²	total_column_cloud_liquid_water	tclw	78	x	x
38	Total column cloud ice water	kg m** ⁻²	total_column_cloud_ice_water	tciw	79	x	x
39	Total column supercooled liquid water	kg m** ⁻²	total_column_supercooled_liquid_water	tcslw	228088		x
40	Total column rain water	kg m** ⁻²	total_column_rain_water	tcrw	228089	x	x
41	Total column snow water	kg m** ⁻²	total_column_snow_water	tcsw	228090	x	x
42	Total column water	kg m** ⁻²	total_column_water	tcw	136	x	x
43	Total column water vapour	kg m** ⁻²	total_column_water_vapour	tcwv	137	x	x
44	Total column ozone	kg m** ⁻²	total_column_ozone	tco3	206	x	x

Table 7: wave parameters: instantaneous

(stream=wave/ewda/wamo/wamd/ewmm/ewmo)
(The native grid is the reduced latitude/longitude grid of 0.36 degrees (1.0 degree for the EDA))

count	name	units	Variable name in CDS	shortName	paramld	an	fc
1	Significant wave height of first swell partition	m	significant_wave_height_of_first_swell_partition	swh1	140121	x	x
2	Mean wave direction of first swell partition	degrees	mean_wave_direction_of_first_swell_partition	mmwd1	140122	x	x
3	Mean wave period of first swell partition	s	mean_wave_period_of_first_swell_partition	mmwp1	140123	x	x
4	Significant wave height of second swell partition	m	significant_wave_height_of_second_swell_partition	swh2	140124	x	x
5	Mean wave direction of second swell partition	degrees	mean_wave_period_of_second_swell_partition	mmwd2	140125	x	x
6	Mean wave period of second swell partition	s	mean_wave_period_of_second_swell_partition	mmwp2	140126	x	x
7	Significant wave height of third swell partition	m	significant_wave_height_of_third_swell_partition	swh3	140127	x	x
8	Mean wave direction of third swell partition	degrees	mean_wave_direction_of_third_swell_partition	mmwd3	140128	x	x
9	Mean wave period of third swell partition	s	mean_wave_period_of_third_swell_partition	mmwp3	140129	x	x
10	Wave Spectral Skewness	dimensionless	wave_spectral_skewness	wss	140207	x	x
11	Free convective velocity over the oceans	m s** ⁻¹	free_convective_velocity_over_the_oceans	wstar	140208	x	x
12	Air density over the oceans	kg m** ⁻³	air_density_over_the_oceans	rhoao	140209	x	x
13	Normalized energy flux into waves	dimensionless	normalized_energy_flux_into_waves	phiaw	140211	x	x
14	Normalized energy flux into ocean	dimensionless	normalized_energy_flux_into_ocean	phioc	140212	x	x

15	Normalized stress into ocean	dimensionless	normalized_stress_into_ocean	tauoc	140214	x	x
16	U-component stokes drift	m s ⁻¹	u_component_stokes_drift	ust	140215	x	x
17	V-component stokes drift	m s ⁻¹	v_component_stokes_drift	vst	140216	x	x
18	Period corresponding to maximum individual wave height	s	period_corresponding_to_maximum_individual_wave_height	tmax	140217	x	x
19	Maximum individual wave height	m	maximum_individual_wave_height	hmax	140218	x	x
20	Model bathymetry	m	model_bathymetry	wmb	140219	x	x
21	Mean wave period based on first moment	s	mean_wave_period_based_on_first_moment	mp1	140220	x	x
22	Mean zero-crossing wave period	s	mean_zero_crossing_wave_period	mp2	140221	x	x
23	Wave spectral directional width	Radians	wave_spectral_directional_width	wdw	140222	x	x
24	Mean wave period based on first moment for wind waves	s	mean_wave_period_based_on_first_moment_for_wind_waves	p1ww	140223	x	x
25	Mean wave period based on second moment for wind waves	s	mean_wave_period_based_on_second_moment_for_wind_waves	p2ww	140224	x	x
26	Wave spectral directional width for wind waves	Radians	wave_spectral_directional_width_for_wind_waves	dwww	140225	x	x
27	Mean wave period based on first moment for swell	s	mean_wave_period_based_on_first_moment_for_swell	p1ps	140226	x	x
28	Mean wave period based on second moment for swell	s	mean_wave_period_based_on_second_moment_for_wind_waves	p2ps	140227	x	x
29	Wave spectral directional width for swell	Radians	wave_spectral_directional_width_for_swell	dwps	140228	x	x
30	Significant height of combined wind waves and swell	m	significant_height_of_combined_wind_waves_and_swell	swh	140229	x	x
31	Mean wave direction	degrees	mean_wave_direction	mwd	140230	x	x
32	Peak wave period	s	peak_wave_period	pp1d	140231	x	x
33	Mean wave period	s	mean_wave_period	mwp	140232	x	x
34	Coefficient of drag with waves	dimensionless	coefficient_of_drag_with_waves	cdww	140233	x	x
35	Significant height of wind waves	m	significant_height_of_wind_waves	shww	140234	x	x
36	Mean direction of wind waves	degrees	mean_direction_of_wind_waves	mdww	140235	x	x
37	Mean period of wind waves	s	mean_period_of_wind_waves	mpww	140236	x	x
38	Significant height of total swell	m	significant_height_of_total_swell	shts	140237	x	x
39	Mean direction of total swell	degrees	mean_direction_of_total_swell	mdts	140238	x	x
40	Mean period of total swell	s	mean_period_of_total_swell	mpts	140239	x	x
41	Mean square slope of waves	dimensionless	mean_square_slope_of_waves	msqs	140244	x	x
42	➤ 10 metre wind speed	m s ⁻¹	ocean_surface_stress_equivalent_10m_neutral_wind_speed	wind	140245	x	x
43	10 metre wind direction	degrees	ocean_surface_stress_equivalent_10m_neutral_wind_direction	dwi	140249	x	x
44	Wave spectral kurtosis	dimensionless	wave_spectral_kurtosis	wsk	140252	x	x
45	Benjamin-Feir index	dimensionless	benjamin_feir_index	bfi	140253	x	x
46	Wave spectral peakedness	dimensionless	wave_spectral_peakedness	wsp	140254	x	x
47	Altimeter wave height	m	Not available from the CDS disks	awh	140246	x	
48	Altimeter corrected wave height	m	Not available from the CDS disks	acwh	140247	x	
49	Altimeter range relative correction	~	Not available from the CDS disks	arrc	140248	x	
50	2D wave spectra (single) ¹	m ² s radian ⁻¹	Not available from the CDS disks	2dfd	140251	x	

¹for 30 frequencies and 24 directions

Table 8: monthly mean surface and single level and wave parameters: exceptions from Tables 1-7

(stream=mnth/moda/edmm/edmo, levtype=sfc or wamo/wamd/ewmm/ewmo)

count	name	units	Variable name in CDS	shortName	paramId	an	fc
1	UV visible albedo for direct radiation	(0 - 1)	uv_visible_albedo_for_direct_radiation	aluvp	15	x	no mean
2	UV visible albedo for diffuse radiation	(0 - 1)	uv_visible_albedo_for_diffuse_radiation	aluvd	16	x	no mean
3	Near IR albedo for direct radiation	(0 - 1)	near_ir_albedo_for_direct_radiation	alinp	17	x	no mean
4	Near IR albedo for diffuse radiation	(0 - 1)	near_ir_albedo_for_diffuse_radiation	alnid	18	x	no mean
5	Magnitude of turbulent surface stress ¹	N m ⁻² s	magnitude of turbulent surface stress	magss	48		x
6	Mean magnitude of turbulent surface stress ²	N m ⁻²	mean magnitude of turbulent surface stress	mmtss	235025		x
7	10 metre wind gust since previous post-processing	m s ⁻¹	10m_wind_gust_since_previous_post_processing	10fg	49		no mean
8	Maximum temperature at 2 metres since previous post-processing	K	maximum_2m_temperature_since_previous_post_processing	mx2t	201		no mean
9	Minimum temperature at 2 metres since previous post-processing	K	minimum_2m_temperature_since_previous_post_processing	mn2t	202		no mean
10	10 metre wind speed ³	m s ⁻¹	10m wind speed	10si	207	x	x
11	Maximum total precipitation rate since previous post-processing	kg m ⁻² s ⁻¹	maximum_total_precipitation_rate_since_previous_post_processing	mxtpr	228226		no mean
12	Minimum total precipitation rate since previous post-processing	kg m ⁻² s ⁻¹	minimum_total_precipitation_rate_since_previous_post_processing	mntpr	228227		no mean
13	Altimeter wave height	m	Not available from the CDS disks	awh	140246	no mean	
14	Altimeter corrected wave height	m	Not available from the CDS disks	acwh	140247	no mean	
15	Altimeter range relative correction	~	Not available from the CDS disks	arrc	140248	no mean	
16	2D wave spectra (single)	m ² s radian ⁻¹	Not available from the CDS disks	2dfd	140251	no mean	

¹Accumulated parameter

²Mean rate/flux parameter

³Instantaneous parameter

Table 9: pressure level parameters: instantaneous

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=pl)

(The native grid is the reduced Gaussian grid N320 (N160 for the EDA) or T639 spherical harmonics (T319 for the EDA), as indicated)

count	name	units	variable name in CDS	shortName	paramId	native grid	an	fc
1	Potential vorticity	K m ² kg ⁻¹ s ⁻¹	potential_vorticity	pv	60	N320 (N160)	x	x
2	Specific rain water content	kg kg ⁻¹	specific_rain_water_content	crwc	75	N320 (N160)	x	x
3	Specific snow water content	kg kg ⁻¹	specific_snow_water_content	cswc	76	N320 (N160)	x	x
4	Geopotential	m ² s ⁻²	geopotential	z	129	T639 (T319)	x	x
5	Temperature	K	temperature	t	130	T639 (T319)	x	x
6	U component of wind	m s ⁻¹	u_component_of_wind	u	131	T639 (T319)	x	x
7	V component of wind	m s ⁻¹	v_component_of_wind	v	132	T639 (T319)	x	x
8	Specific humidity	kg kg ⁻¹	specific_humidity	q	133	N320 (N160)	x	x
9	Vertical velocity	Pa s ⁻¹	vertical_velocity	w	135	T639 (T319)	x	x
10	Vorticity (relative)	s ⁻¹	vorticity	vo	138	T639 (T319)	x	x

11	Divergence	s**-1	divergence	d	155	T639 (T319)	x	x
12	Relative humidity	%	relative_humidity	r	157	T639 (T319)	x	x
13	Ozone mass mixing ratio	kg kg**-1	ozone_mass_mixing_ratio	o3	203	N320 (N160)	x	x
14	Specific cloud liquid water content	kg kg**-1	specific_cloud_liquid_water_content	clwc	246	N320 (N160)	x	x
15	Specific cloud ice water content	kg kg**-1	specific_cloud_ice_water_content	ciwc	247	N320 (N160)	x	x
16	Fraction of cloud cover	(0 - 1)	fraction_of_cloud_cover	cc	248	N320 (N160)	x	x

Table 10: potential temperature level parameters: [instantaneous](#)

(not available from the CDS disks)

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=pt)

(The native grid is the reduced Gaussian grid N320 (N160 for the EDA) or T639 spherical harmonics (T319 for the EDA), as indicated)

count	name	units	shortName	paramId	native grid	an	fc
1	Montgomery potential	m**2 s**-2	mont	53	T639 (T319)	x	
2	Pressure	Pa	pres	54	T639 (T319)	x	
3	Potential vorticity	K m**2 kg**-1 s**-1	pv	60	N320 (N160)	x	
4	U component of wind	m s**-1	u	131	T639 (T319)	x	
5	V component of wind	m s**-1	v	132	T639 (T319)	x	
6	Specific humidity	kg kg**-1	q	133	N320 (N160)	x	
7	Vorticity (relative)	s**-1	vo	138	T639 (T319)	x	
8	Divergence	s**-1	d	155	T639 (T319)	x	
9	Ozone mass mixing ratio	kg kg**-1	o3	203	N320 (N160)	x	

Table 11: potential vorticity level parameters: [instantaneous](#)

(not available from the CDS disks)

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=pv)

(The native grid is the reduced Gaussian grid N320 (N160 for the EDA) or T639 spherical harmonics (T319 for the EDA), as indicated)

count	name	units	shortName	paramId	native grid	an	fc
1	Potential temperature	K	pt	3	T639 (T319)	x	
2	Pressure	Pa	pres	54	T639 (T319)	x	
3	Geopotential	m**2 s**-2	z	129	T639 (T319)	x	
4	U component of wind	m s**-1	u	131	N320 (N160)	x	
5	V component of wind	m s**-1	v	132	N320 (N160)	x	
6	Specific humidity	kg kg**-1	q	133	N320 (N160)	x	
7	Ozone mass mixing ratio	kg kg**-1	o3	203	N320 (N160)	x	

Table 12: model level parameters: [instantaneous](#)

(GRIB2 format)

(not available from the CDS disks)

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=ml)

(The native grid is the reduced Gaussian grid N320 (N160 for the EDA) or T639 spherical harmonics (T319 for the EDA), as indicated)

count	name	units	shortName	paramId	native grid	an	fc
1	Specific rain water content	kg kg**-1	crwc	75	N320 (N160)	x	x
2	Specific snow water content	kg kg**-1	cswc	76	N320 (N160)	x	x
3	Eta-coordinate vertical velocity	s**-1	etadot	77	T639 (T319)	x	x
4	Geopotential ¹	m**2 s**-2	z	129	T639 (T319)	x	x
5	Temperature	K	t	130	T639 (T319)	x	x
6	U component of wind	m s**-1	u	131	T639 (T319)	x	x
7	V component of wind	m s**-1	v	132	T639 (T319)	x	x
8	Specific humidity	kg kg**-1	q	133	N320 (N160)	x	x
9	Vertical velocity	Pa s**-1	w	135	T639 (T319)	x	x
10	Vorticity (relative)	s**-1	vo	138	T639 (T319)	x	x
11	Logarithm of surface pressure ¹	~	Insp	152	T639 (T319)	x	x
12	Divergence	s**-1	d	155	T639 (T319)	x	x
13	Ozone mass mixing ratio	kg kg**-1	o3	203	N320 (N160)	x	x
14	Specific cloud liquid water content	kg kg**-1	clwc	246	N320 (N160)	x	x
15	Specific cloud ice water content	kg kg**-1	ciwc	247	N320 (N160)	x	x
16	Fraction of cloud cover	(0 - 1)	cc	248	N320 (N160)	x	x

¹Only archived on level=1.

Table 13: model level parameters: [mean rates/fluxes](#)

(GRIB2 format)

(not available from the CDS disks)

(stream=oper/enda/mnth/moda/edmm/edmo, levtype=ml)

(The native grid is the reduced Gaussian grid N320 (N160 for the EDA))

count	name	units	shortName	paramId	an	fc
1	Mean temperature tendency due to short-wave radiation	K s**-1	mttswr	235001		x
2	Mean temperature tendency due to long-wave radiation	K s**-1	mttlwr	235002		x
3	Mean temperature tendency due to short-wave radiation, clear sky	K s**-1	mttswrCS	235003		x
4	Mean temperature tendency due to long-wave radiation, clear sky	K s**-1	mttlwrCS	235004		x
5	Mean temperature tendency due to parametrisations	K s**-1	mttprm	235005		x
6	Mean specific humidity tendency due to parametrisations	kg kg**-1 s**-1	mqtpm	235006		x
7	Mean eastward wind tendency due to parametrisations	m s**-2	mutprm	235007		x
8	Mean northward wind tendency due to parametrisations	m s**-2	mvtpm	235008		x
9	Mean updraught mass flux¹	kg m**-2 s**-1	mumf	235009		x
10	Mean downdraught mass flux¹	kg m**-2 s**-1	mdmf	235010		x
11	Mean updraught detrainment rate	kg m**-3 s**-1	mudr	235011		x
12	Mean downdraught detrainment rate	kg m**-3 s**-1	mddr	235012		x
13	Mean total precipitation flux¹	kg m**-2 s**-1	mtpf	235013		x
14	Mean turbulent diffusion coefficient for heat¹	m**2 s**-1	mtdch	235014		x

¹These parameters provide data for the model half levels - the interfaces of the model layers.

Observations

The observations (satellite and in-situ) used as input to ERA5 are listed below. For more information on the observational input to ERA5, including dates when particular sensors or observation types were used, please see Section 5 in the ERA5 journal article, [The ERA5 global reanalysis](#).

- [Table 14: Satellite Data](#)
- [Table 15: In-situ data, provided by WMO WIS](#)

Table 14: Satellite Data

Sensor	Satellite	Satellite agency	Data provider+	Measurement (sensitivities exploited in ERA5 / variables analysed)
Satellite radiances (infrared and microwave)				
AIRS	AQUA	NASA	NOAA	BT (T, humidity and ozone)
AMSR-2	GCOM-W1*	JAXA		BT (column water vapour, cloud liquid water, precipitation and ocean surface wind speed)
AMSRE	AQUA*	JAXA		BT (column water vapour, cloud liquid water, precipitation and ocean surface wind speed)
AMSUA	NOAA-15/16/17/18/19, AQUA, METOP-A/B	NOAA,ESA,EUMETSAT		BT (T)
AMSUB	NOAA-15/16/17	NOAA		BT (humidity)
ATMS	NPP	NOAA		BT (T and humidity)
CRIS	NPP	NOAA		BT (T, humidity and ozone)
HIRS	TIROS-N, NOAA-6 /7/8/9/11/14	NOAA		BT (T, humidity and ozone)
IASI	METOP-A/B	EUMETSAT/ESA	EUMETSAT	BT (T, humidity and ozone)
GMI	GPM	NASA/JAXA		BT (humidity, column water vapour, cloud liquid water, precipitation, ocean surface wind speed)
MHS	NOAA-18/19, METOP-A/B	NOAA, EUMETSAT/ESA		BT (humidity and precipitation)
MSU	TIROS-N, NOAA-6 to 12, NOAA-14			BT (T)
MWHS	FY-3-A/B	NRSCC		BT (humidity)
MWHS2	FY-3-C	CMA		BT (T, humidity and precipitation)
MWTS	FY-3A/B	NRSCC		BT (T)
MWTS2	FY-3C	CMA		BT (T)
SSM/I	DMSP-08*/10*/11*/13*/14*/15*	US Navy	NOAA,CMSAF*	BT (column water vapour, cloud liquid water, precipitation and ocean surface wind speed)
SSMIS	DMSP-16/17/18	US Navy	NOAA	BT (T, humidity, column water vapour, cloud liquid water, precipitation and ocean surface wind speed)
SSU	TIROS-N, NOAA-6/7/8/9/11/14	NOAA		BT (T)
TMI	TRMM	NASA/JAXA		BT (column water vapour, cloud liquid water, precipitation, ocean surface wind speed)
MVIRI	METEOSAT 5/7	EUMETSAT/ESA	EUMETSAT	BT (water vapour, surface/cloud top T)
SEVIRI	METEOSAT-8*/9*/10	EUMETSAT/ESA	EUMETSAT	BT (water vapour, surface/cloud top T)
GOES IMAGER	GOES-8/9/10/11/12/13/15	NOAA	CIMMS,NESDIS	BT (water vapour, surface/cloud top T)
MTSAT IMAGER	MTSAT-1R/MTSAT-2	JMA		BT (water vapour, surface/cloud top T)
AHI	Himawari-8	JMA		BT (water vapour, surface/cloud top T)
Satellite retrievals from radiance data				
MVIRI	METEOSAT-2*/3*/4*/5*/7*	EUMETSAT/ESA	EUMETSAT	wind vector
SEVIRI	METEOSAT-8*/9*/10	EUMETSAT/ESA	EUMETSAT	wind vector
GOES IMAGER	GOES-4-6/8*/9*/10*/11*/12*/13*/15*	NOAA	CIMMS*,NESDIS	wind vector
GMS IMAGER	GMS-1*/2/3*/4*/5*	JMA		wind vector
MTSAT IMAGER	MTSAT-1R*/MTSAT2	JMA		wind vector
AHI	Himawari-8	JMA	JMA	wind vector
AVHRR	NOAA-7 /9/10/11/12/14 to 18, METOP-A	NOAA	CIMMS,EUMETSAT	wind vector
MODIS	AQUA/TERRA	NASA	NESDIS,CIMMS	wind vector
GOME	ERS-2*	ESA		Ozone
GOME-2	METOP*-A/B	ESA/EUMETSAT		Ozone
MIPAS	ENVISAT*	ESA		Ozone
MLS	EOS-AURA*	NASA		Ozone
OMI	EOS-AURA*	NASA		Ozone
SBUV,SBUV-2	NIMBUS-7*,NOAA*9/11/14/16/17/18/19	NOAA	NASA	Ozone
SCIAMACHY	ENVISAT*	ESA		Ozone
TOMS	NIMBUS-7*,METEOR-3-5,ADEOS-1*,EARTH PROBE	NASA		Ozone
Satellite GPS-Radio Occultation data				
BlackJack	CHAMP,GRACE*-A/B,SAC-C*	DLR,NASA/DLR,NASA/COMAE	GFZ,UCAR*	Bending angle
GRAS	METOP-A/B	EUMETSAT/ESA	EUMETSAT	Bending angle
IGOR	TerraSAR-X*, TanDEM-X, COSMIC*-1 to 6	NSPO/NOAA	GFZ,UCAR*	Bending angle
Satellite scatterometer data				
AMI	ERS-1,ERS-2	ESA		Backscatter sigma0, soil moisture
ASCAT	METOP-A/B*	EUMETSAT/ESA	EUMETSAT/TU Wien	Backscatter sigma0, soil moisture
OSCAT	OCEANSAT-2	ISRO	KNMI	Backscatter sigma0, vector wind
SEAWINDS	QUIKSCAT	NASA	NASA	Backscatter sigma0
Satellite Altimeter data				
RA	ERS-1*/2*	ESA		Wave Height
RA-2	ENVISAT*	ESA		Wave Height
Poseidon-2	JASON-1*	CNES/NASA	CNES	Wave Height
Poseidon-3	JASON-2	CNES/NOAA/NASA/EUMETSAT	NOAA/EUMETSAT	Wave Height
SIRAL	CRYOSAT-2	ESA		Wave Height
AltiKa	SARAL	CNES/ISRO	EUMETSAT	Wave Height

* reprocessed dataset
+ when different than the satellite agency

Table 15: In-situ data, provided by WMO WIS

Dataset name	Observation type	Measurement
SYNOP	Land station	Surface Pressure, Temperature, humidity
METAR	Land station	Surface Pressure, Temperature, humidity
DRIBU/DRIBU-BATHY/DRIBU-TESAC/BUFR Drifting Buoy	Drifting buoys	10m-wind, Surface Pressure
BUFR Moored Buoy	Moored buoys	10m-wind, Surface Pressure
SHIP	ship station	Surface Pressure, Temperature, wind, humidity
Land/ship PILOT	Radiosondes	wind profiles
American Wind Profiler	Radar	wind profiles

European Wind Profiler	Radar	wind profiles
Japanese Wind Profiler	Radar	wind profiles
TEMP SHIP	Radiosondes	Temperature, wind, humidity profiles
DROP Sonde	Radiosondes	Temperature, wind, humidity profiles
Land/Mobile TEMP	Radiosondes	Temperature, wind, humidity profiles
AIREP	Aircraft data	Temperature, wind
AMDAR	Aircraft data	Temperature, wind
ACARS	Aircraft data	Temperature, wind, humidity
WIGOS AMDAR	Aircraft data	Temperature, wind, humidity
TAMDAR	Aircraft data	Temperature, wind
ADS-C	Aircraft data	Temperature, wind
Mode-S	Aircraft data	Wind
Ground based radar	Radar precipitation composites	Rain rates

Table 16: Snow data

Dataset name	Observation type	Measurement
SYNOP	Land station	Snow depth
Additional national reports	Land station	Snow depth
NOAA/NESDIS IMS	Merged satellite	Snow cover (NH only)

Guidelines

The following advice is intended to help users understand particular features of the ERA5 data:

- In general, we recommend that the hourly (analysed) "2 metre temperature" be used to construct the minimum and maximum over longer periods, such as a day, rather than using the forecast parameters "Maximum temperature at 2 metres since previous post-processing" and "Minimum temperature at 2 metres since previous post-processing".
- ERA5: compute pressure and geopotential on model levels, geopotential height and geometric height
- ERA5: How to calculate wind speed and wind direction from u and v components of the wind?
- Sea surface temperature and sea-ice cover (sea ice area fraction), see Table 2 above, are available at the usual times, eg hourly for the HRES, but their content is only updated once daily. However, for inland water bodies (lakes, reservoirs, rivers and coastal waters) the FLake model calculates the surface temperature (ie the lake mixed-layer temperature or lake ice temperature) and does include diurnal variations.
- Mean rates/fluxes and accumulations at step=0 have values of zero because the length of the processing period is zero.
- Convective Inhibition (CIN). A missing value is assigned to CIN for values of CIN > 1000 or where there is no cloud base. This can occur where convective available potential energy (CAPE) is low.
- ERA5: mixing CDS and MARS data
- ERA5: Land-sea mask for wave variables
- Altimeter wave parameters
- Computation of near-surface humidity
- Computation of snow cover
- "Forecast albedo" is only for diffuse radiation
- Actual and potential evapotranspiration
- Values below the surface are there for convenience and simplicity: the fields would be more difficult to deal with if they contained voids. This can be problematic for many applications, such as time averages where the voids move around with time. However, extrapolating into the data voids can contaminate the "real" atmosphere. This happens if the parameter is archived in spectral space or if it's horizontally interpolated. The method for extrapolation below the surface is parameter dependent, but wind components and humidity are kept constant from the lowest model level, whereas temperature is linearly interpolated from the lowest model level to a surface temperature and then a cubic polynomial extrapolation is used thereafter.
- "Evaporation" and "Instantaneous moisture flux"

Known issues

Currently, we are aware of these issues with ERA5:

- ERA5T: from 1 September to 13 December 2021, the final ERA5 product is different to ERA5T due to the correction of the assimilation of incorrect snow observations in central Asia. Although the differences are mostly limited to that region and mainly to surface parameters, in particular snow depth and soil moisture and to a lesser extent 2m temperature and 2m dewpoint temperature, all the resulting reanalysis fields can differ over the whole globe but should be within their range of uncertainty (which is estimated by the ensemble spread and which can be large for some parameters). On the CDS disks, the initial ERA5T fields have been overwritten (with the usual 2-3 month delay), i.e., for these months, access to the original CDS disk, ERA5T product is not possible after it has been overwritten. Potentially incorrect snow observations have been assimilated in ERA5 up to this time, when the effects became noticeable. The quality control of snow observations has been improved in ERA5 from September 2021 and from 15 November 2021 in ERA5T.
- ERA5 uncertainty: although small values of ensemble spread correctly mark more confident estimates than large values, numerical values are over confident. The spread does give an indication of the relative, random uncertainty in space and time.
- ERA5 suffers from an overly strong equatorial mesospheric jet, particularly in the transition seasons.
- From 2000 to 2006, ERA5 has a poor fit to radiosonde temperatures in the stratosphere, with a cold bias in the lower stratosphere. In addition, a warm bias higher up persists for much of the ERA5 period. The lower stratospheric cold bias was rectified in a re-run for the years 2000 to 2006, called ERA5.1, see "Resolved issues" below.
- Discontinuities in ERA5: The historic ERA5 data was produced by running several parallel experiments, each for a different period, which were then spliced together to create the final product. This can create discontinuities at the transition points.
- The analysed "2 metre temperature" can be larger than the forecast "Maximum temperature at 2 metres since previous post-processing".
- The analysed 10 metre wind speed (derived from the 10 metre wind components) can be larger than the forecast "10 metre wind gust since previous post-processing".
- ERA5 diurnal cycle for near surface winds: the hourly data reveals a mismatch in the analysed near surface wind speed between the end of one assimilation cycle and the beginning of the next (which occurs at 9:00 - 10:00 and 21:00 - 22:00 UTC). This problem mostly occurs in low latitude oceanic regions, though it can also be seen over Europe and the USA. We cannot rectify this problem in the analyses. The forecast near surface winds show much better agreement between the assimilation cycles, at least on average, so if this mismatch is problematic for a particular application, our advice would be to use the forecast winds. The forecast near surface winds are available from MARS, see the section [Data organisation and how to download ERA5](#).
- ERA5 diurnal cycle for near surface temperature and humidity: some locations do suffer from a mismatch in the analysed values between the end of one assimilation cycle and the beginning of the next, in a similar fashion to that for the near surface winds (see above), but this problem is thought not to be so widespread as that for the near surface winds. The forecast values for near surface temperature and humidity are usually smoother than the analyses, but the forecast low level temperatures suffer from a cold bias over most parts of the globe. The forecast near surface temperature and humidity are available from MARS, see the section [Data organisation and how to download ERA5](#).
- ERA5: large 10m winds: up to a few times per year, the analysed low level winds, eg 10m winds, become very large in a particular location, which varies amongst a few apparently preferred locations. The largest values seen so far are about 300 ms⁻¹.
- ERA5 rain bombs: up to a few times per year, the rainfall (precipitation) can become extremely large in small areas. This problem occurs mostly over Africa, in regions of high orography.
- Large values of CAPE: occasionally, the Convective available potential energy in ERA5 is unrealistically large.
- Ship tracks in the SST: prior to September 2007, in the period when HadISST2 was used, ship tracks can be visible in the SST.
- Prior to 2014, the SST was not used over the Great Lakes to nudge the lake model. Consequently, the 2 metre temperature has an annual cycle that is too strong, with temperatures being too cold in winter and too warm in summer.
- The Potential Evaporation field (pev, parameter id 228251) is largely underestimated over deserts and high-forested areas. This is due to a bug in the code that does not allow transpiration to occur in the situation where there is no low vegetation.
- Wave parameters (Table 7 above) for the three swell partitions: these parameters have been calculated incorrectly. The problem is most evident in the swell partition parameters involving the mean wave period: Mean wave period of first swell partition, Mean wave period of second swell partition and Mean wave period of third swell partition, where the periods are far too long.
- Surface photosynthetically available radiation (PAR) is too low in the version (CY41R2) of the ECMWF Integrated Forecasting System (IFS) used to produce ERA5, so PAR and clear sky PAR have not been published in ERA5. There is a bug in the calculation of PAR, with it being taken from the wrong parts of the spectrum. The shortwave bands include 0.442-0.625 micron, 0.625-0.778 micron and 0.778-1.24 micron. PAR should be coded to be the sum of the radiation in the first of these bands and 0.42 of the second (to account for the fact that PAR is normally defined to stop at 0.7 microns). However, in CY41R2, PAR is in fact calculated from the sum of the second band plus 0.42 of the third. We will try to fix this in a future cycle.
- The instantaneous turbulent surface stress components (eastward and northward) and friction velocity tend to be too small
- ERA5 forecast parameters are missing for the validity times of 1st January 1940 from 00 UTC to 06 UTC (except for forecast step=0). This problem occurs because the first forecast in ERA5 was initiated from 1st January 1940 at 06 UTC.
- Maximum temperature at 2 metres since previous post-processing: in a small region over Peru, at 19 UTC, 2 August 2013, this forecast parameter exhibited erroneous values, which were greater than 50C. This occurrence is under investigation. Note, in general, we recommend that the hourly (analysed) "2 metre temperature" be used to construct the minimum and maximum over longer periods, such as a day.
- Four reasons why hourly data might not be consistent with their monthly mean
- ERA5 sea-ice cover and 2 metre temperature: in the period 1979-1989, in a region just to the north of Greenland, the sea-ice cover outside of the melt season is too low and hence the 2 metre temperature is too high. For more information, see Section 3.5.4 of [Low frequency variability and trends in surface air temperature and humidity from ERA5 and other datasets](#)
- ERA5 sea-ice cover is missing in the Caspian Sea from late 2007 to 2013, inclusive.
- ERA5 sea-ice surface temperature (skin temperature) in the Arctic, during winter, can have a warm bias of 5K or more. This issue is most pronounced over thick snow-covered sea ice under cold clear-sky conditions, when the modelled conductive heat flux from the warm ocean underneath the ice and snow layer is too high. More information can be found in [Batra](#) and [Müller](#) (2019) and [Zampieri et al., \(2023\)](#), the latter of which, also describes a method to improve on this bias.
- Altimeter wave height observations have not been available for ERA5 in the following periods (since coverage began in mid-1991): early February 2021 to mid-January 2022; mid-October 2023 onwards.
- ERA5 CDS: wind values are far too low on pressure levels at the poles in the CDS
- Snow present in Iberia throughout 1978 due to assimilation of erroneous in situ snow data. This has an effect on 2m temperature, which shows negative anomalies of several degrees Celsius. There is no snow present in ERA5-Land, as this snow data is not assimilated. However, the 2m temperature anomaly is present, as the forcing comes from the erroneous ERA5 data. These figures show ERA monthly averaged 2m temperature (t2m) and snow depth (sd) (38 to 43N, -8 to -6W), from 1940-2023, with 1978 highlighted in red. The same snow depth plot, limited to 1977-07 to 1979-04 shows more detail, with the period of erroneous snow depth in ERA5 extending from 1977-12 to 1979-03 in the [monthly mean dataset](#).
- Constant SST values from 24th to 29th June 2024
- Constant SIC values from 11th to 18th September 2024

Resolved issues

- ERA5.1 is a re-run of ERA5, for the years 2000 to 2006 only, and was produced to improve upon the cold bias in the lower stratosphere seen in ERA5.
 - [More information and details for downloading ERA5.1](#)
- ERA5.1 CDS: If you retrieved ERA5.1 using the CDS API anytime before 20/05/2020 08:00 UTC, for any stream other than oper (i.e. streams: wave, enda, edmo, ewmo, edmm, ewmm, ewda, moda, wamd, mnth, wamo), you will need to request the data again. Prior to this date, stream oper would be delivered regardless of which stream was requested.
- ERA5 CDS: incorrect values of QJv on pressure levels in the CDS

4. [ERA5 CDS: Data corruption](#)
5. **ERA5T rewritten:** from **1 July to 17 November 2024, the final ERA5 product was different to ERA5T** due to the correction of [the assimilation of incorrect snow observations on the Alps](#). Although the differences were mostly limited to that region and mainly to surface parameters, in particular snow depth and soil moisture and to a lesser extent 2m temperature and 2m dewpoint temperature, all the resulting reanalysis fields can differ over the whole globe but should be within their range of uncertainty (which is estimated by the ensemble spread and which can be large for some parameters). On the CDS disks, the initial, ERA5T, fields have been overwritten (with the usual 2-3 month delay), i.e., for these months, access to the original CDS disk, ERA5T product is not possible after it has been overwritten. Potentially incorrect snow observations have been assimilated in ERA5 up to this time, when the effects became noticeable.
6. **ERA5T rewritten 1st - 6th January 2025:** At the turn of 2025, an unfortunate technical issue with the production of ERA5 resulted in a large part of satellite observations not being used. This led to small but systematic differences, impacting mainly humidity in the tropics, and in lesser extent other ERA5 parameters as well from the 1st of January 2025 onwards. The issue was resolved and ERA5T and ERA5-Land-T were rewritten with the correct data from 12 UTC 1st January to 12 UTC 6th January 2025. The consolidated ERA5 and ERA5-Land will contain these corrected values.

User support

There is a range of user support available for ERA5, including a Knowledge Base (where this article resides), a Forum and a ticketed system for questions - for more information see the [C3S Help and Support Page](#).

How to acknowledge and cite ERA5

If you have downloaded ERA5 data on the "CDS disks" and/or downloaded ERA5 data in MARS, using either the CDS API ('`reanalysis-era5-complete`' or '`reanalysis-era5.1-complete`') or via authorised direct access to MARS, please follow the instructions below:

In addition to the terms and conditions of the license(s), users **must**:

- cite the CDS catalogue entry;
- provide clear and visible attribution to the Copernicus programme and attribute each data product used;

Step 1: Check the licence for any attribution/reference clause

Step 2: Cite the CDS catalogue entry (as traceable source of data). Note that a catalogue entry for ERA5-complete and ERA5.1 is now also available in the CDS.

Step 3: Provide clear and visible attribution to the Copernicus programme and attribute each data product used (to accredit the creators of the data). Throughout the content of your publication, the dataset used is referred to as Author (YYYY)

The 3-steps procedure above is illustrated with this example: [Use Case 2: ERA5 hourly data on single levels from 1940 to present](#)

For complete details, please refer to [How to acknowledge and cite a Climate Data Store \(CDS\) catalogue entry and the data published as part of it](#).

References

[The ERA5 global reanalysis](#)

[The ERA5 global reanalysis from 1940 to 2022](#)

[The ERA5 global reanalysis: Preliminary extension to 1950](#)

[Global stratospheric temperature bias and other stratospheric aspects of ERA5 and ERA5.1](#)

[Low frequency variability and trends in surface air temperature and humidity from ERA5 and other datasets](#)

Further ERA5 references are available from the [ECMWF website](#).

This document has been produced in the context of the Copernicus Climate Change Service (C3S).

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